



REPORT

**Design and Development of a Stationery E-Commerce Website
Using HTML, CSS & JavaScript**

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Abstract

This project report presents the design and development of a **Stationery E-Commerce Website** using modern web technologies including **HTML, CSS, and JavaScript**. The website allows users to view stationery products such as pens, notebooks, and highlighters, while enabling the administrator to manage products through **CRUD (Create, Read, Update, Delete) operations**. The project demonstrates the practical application of ICT concepts, front-end development, and client-side data management using **Local Storage**. The system is designed to be simple, user-friendly, and suitable for small-scale businesses.

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1. Introduction

1.1 Background

Information and Communication Technology has transformed traditional businesses into digital platforms. E-commerce websites enable customers to browse products, compare prices, and make informed purchasing decisions. Stationery products are essential in educational institutions and offices, making them suitable for online selling.

1.2 Project Overview

This project focuses on developing a **Stationery E-Commerce Website** that displays various stationery items and allows administrators to manage product records dynamically. The website is fully client-side and does not require a backend server

2. Project Objectives

- To design a visually appealing stationery store website
- To implement CRUD operations using JavaScript
- To understand Local Storage-based data handling
- To apply ICT concepts practically
- To enhance front-end development skills

3. Problem Statement

Many small stationery shops lack an online presence due to high development costs. This project aims to provide a low-cost, easy-to-maintain e-commerce solution using basic web technologies.

4. Scope of the Project

In-Scope Features

- Display stationery products
- Add, update, and delete products
- Store product data locally

Out-of-Scope Features:

- Online payments
- User login system
- Order tracking

5. SDLC Methodology

5.1 Selected Model: Waterfall Model

PHASE	DESCRIPTION
Requirement Analysis	Identify product management needs
System Design	Design layout and navigation
Implementation	Develop using HTML, CSS, JS
Testing	Validate CRUD operations
Deployment	Run on browser

6. Tools & Technologies

Technology	Purpose
HTML	Page Structure
CSS	Styling and Layout
JavaScript	CRUD Operations
Local Storage	Data Storage
VS code	Development
Browser	Execution

7. System Requirements

7.1 Hardware Requirements

Component	Specifications
Processor	Any Modern CPU
RAM	Minimum 4GB
Storage	500 MB

7.2 Software Requirements

Software	Version
Operating System	Windows
Browser	Chrome
Code Editor	VS Code

8. System Design

8.1 Website Architecture

The system follows a client-side architecture where all processing and data storage occur within the browser.

8.2 Navigation Flow

Home → Stationery Products → Add / Update / Delete Product → About us/Contact us

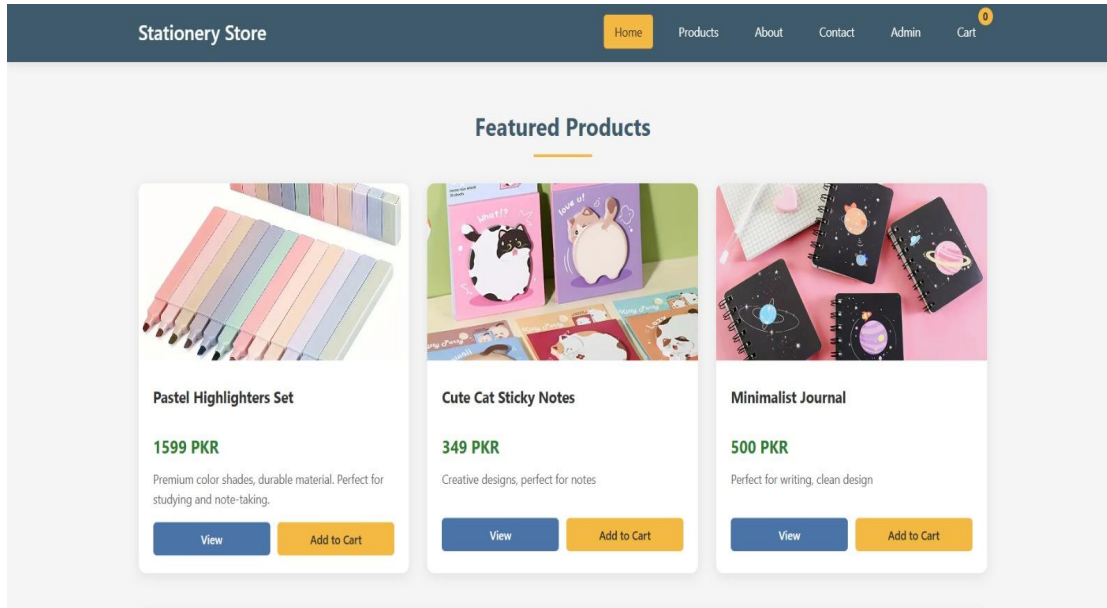
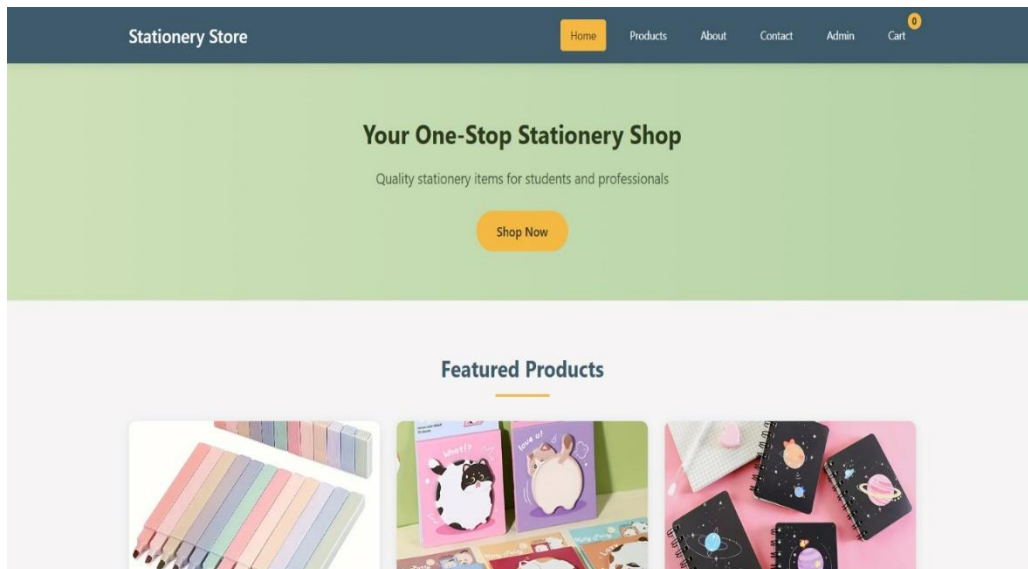
9. Data Handling Design

Field	Description
Product ID	Unique identifier
Product Name	Name of stationery item
Price	Product price
Quantity	Available stock
Description	Product details

10. Implementation & Screenshots

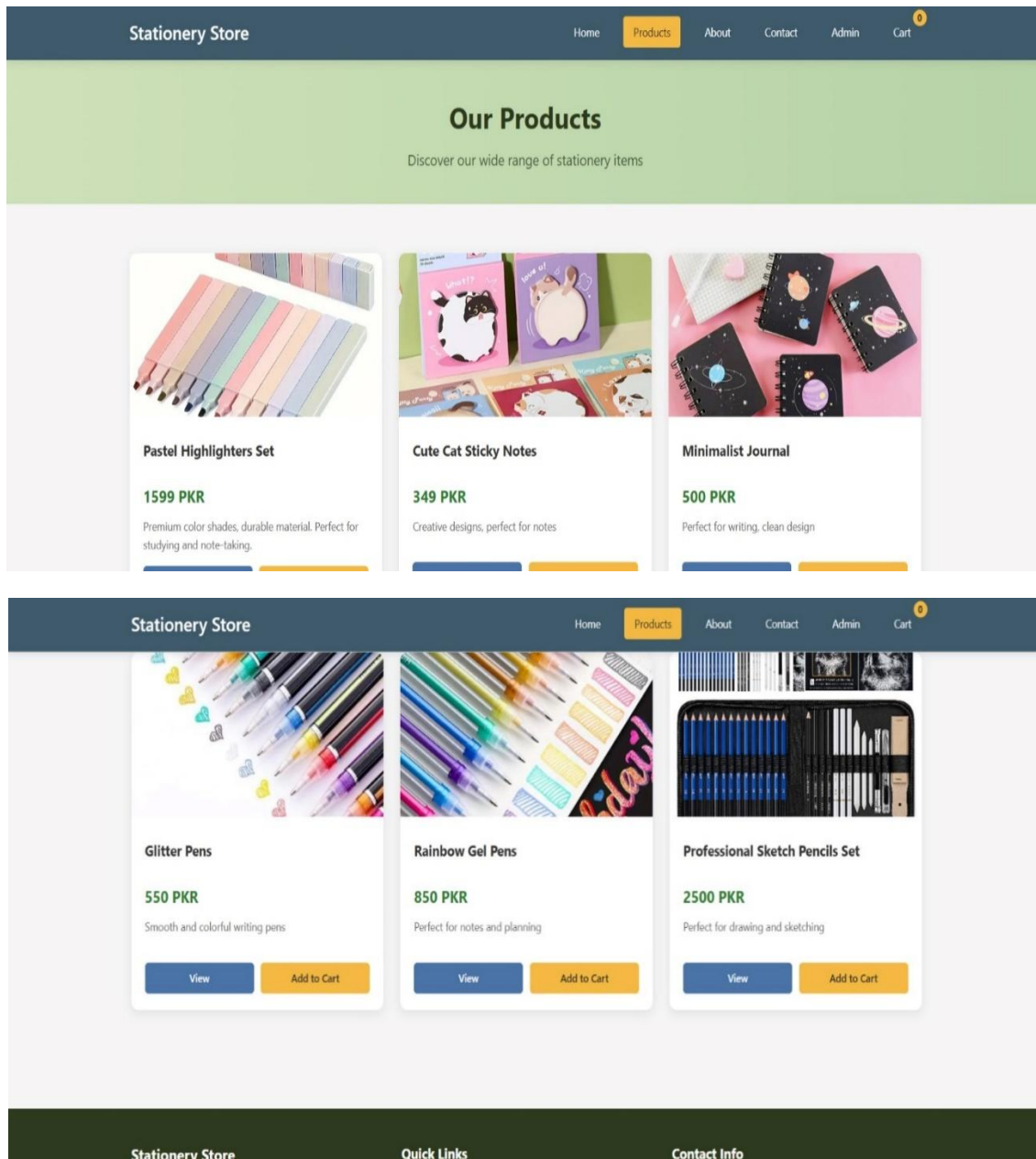
10.1 Home Page

Description: Displays website title, navigation bar, and introductory text.



10.2 Product Page

Description: Displays the list of stationery products with images, prices, and brief descriptions. Users can view product details, select quantity, and use action buttons such as **Add to Cart** or **Buy Now** for purchasing items.



10.2 Add Product Page (CREATE)

Description: Admin can add new stationery items using this form.

The screenshot shows the 'Stationery Store' Admin Panel. At the top, there's a navigation bar with 'Home', 'Products', 'About', 'Contact', 'Admin' (highlighted), and 'Cart' (with a notification badge). The main content area is titled 'Admin Panel'. It contains a form for adding a new product with fields for 'Product Name', 'Price (PKR)', and 'Description'. Below the form is an orange button labeled '+ Add Product'. Underneath the form is a 'Product List' table. The table has columns for ID, Name, Price, Description, and Actions. The first row shows a product with ID 1, Name 'Pastel Highlighters Set', Price '1599 PKR', and Description 'Premium color shades, durable material. Perfect for studying and note-taking.' The Actions column for this row has an 'Edit' button.

ID	Name	Price	Description	Actions
1	Pastel Highlighters Set	1599 PKR	Premium color shades, durable material. Perfect for studying and note-taking.	<button>Edit</button>

10.3 Stationery Product Listing Page (READ)

Description: Shows all stationery items dynamically fetched from Local Storage

The screenshot shows the 'Stationery Store' Product Listing Page. It features a navigation bar at the top with 'Home', 'Products', 'About', 'Contact', 'Admin' (highlighted), and 'Cart' (with a notification badge). The main content area displays a table of stationery products. The table has columns for ID, Name, Price, Description, and Actions. There are six rows of products, each with an 'Edit' and 'Delete' button in the Actions column.

ID	Name	Price	Description	Actions
1	Pastel Highlighters Set	1599 PKR	Premium color shades, durable material. Perfect for studying and note-taking.	<button>Edit</button> <button>Delete</button>
2	Cute Cat Sticky Notes	349 PKR	Creative designs, perfect for notes	<button>Edit</button> <button>Delete</button>
3	Minimalist Journal	500 PKR	Perfect for writing, clean design	<button>Edit</button> <button>Delete</button>
4	Glitter Pens	550 PKR	Smooth and colorful writing pens	<button>Edit</button> <button>Delete</button>
5	Rainbow Gel Pens	850 PKR	Perfect for notes and planning	<button>Edit</button> <button>Delete</button>
6	Professional Sketch Pencils Set	2500 PKR	Perfect for drawing and sketching	<button>Edit</button> <button>Delete</button>

10.4 Update Product Page (UPDATE)

Description: Allows editing existing stationery product details

Stationery Store					Home	Products	About	Contact	Admin	Cart ³
ID	Name	Price	Description	Actions						
1	Pastel Highlighters Set	1599 PKR	Premium color shades, durable material. Perfect for studying and note-taking.	Edit	Delete					
2	Cute Cat Sticky Notes	349 PKR	Creative designs, perfect for notes	Edit	Delete					
3	Minimalist Journal	500 PKR	Perfect for writing, clean design	Edit	Delete					
4	Glitter Pens	550 PKR	Smooth and colorful writing pens	Edit	Delete					
5	Rainbow Gel Pens	850 PKR	Perfect for notes and planning	Edit	Delete					
6	Professional Sketch Pencils Set	2500 PKR	Perfect for drawing and sketching	Edit	Delete					

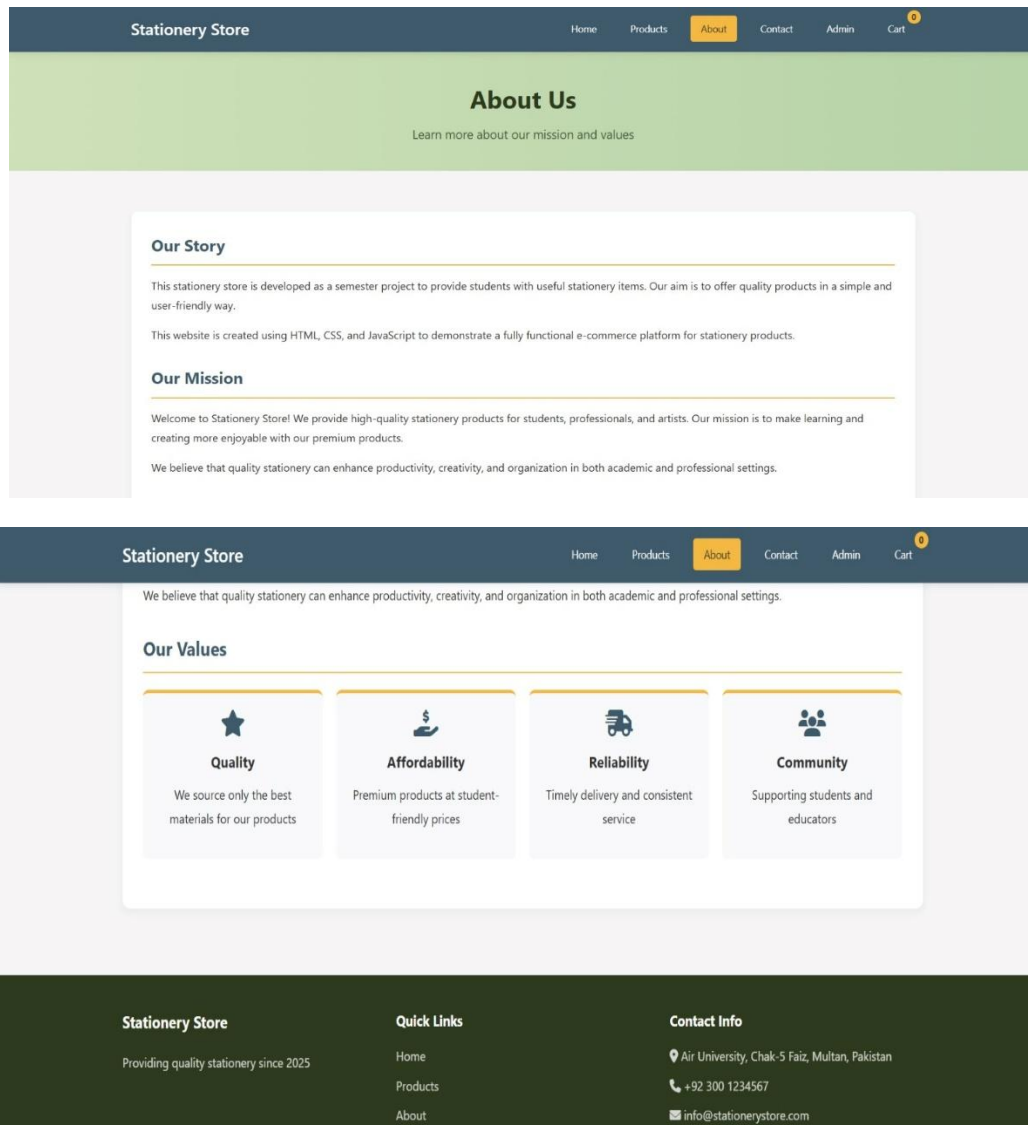
10.5 Delete Product Functionality (DELETE)

Description: Removes selected stationery products from the system

Stationery Store					Home	Products	About	Contact	Admin	Cart ³
ID	Name	Price	Description	Actions						
1	Pastel Highlighters Set	1599 PKR	Premium color shades, durable material. Perfect for studying and note-taking.	Edit	Delete					
2	Cute Cat Sticky Notes	349 PKR	Creative designs, perfect for notes	Edit	Delete					
3	Minimalist Journal	500 PKR	Perfect for writing, clean design	Edit	Delete					
4	Glitter Pens	550 PKR	Smooth and colorful writing pens	Edit	Delete					
5	Rainbow Gel Pens	850 PKR	Perfect for notes and planning	Edit	Delete					
6	Professional Sketch Pencils Set	2500 PKR	Perfect for drawing and sketching	Edit	Delete					

10.6 About Us Page

Description: Provides information about the stationery store, including its purpose, mission, and services offered. This page helps users understand the background of the website and builds trust with customers.



10.4 Contact Us Page

Description: Displays contact details and a user inquiry form. Users can enter their name, email address, and message to communicate with the store. This page enables customer support and feedback submission.

Stationery Store Home Products About **Contact** Admin Cart 0

Contact Us

Get in touch with our team

Get In Touch

Address
Air University, Chak-S Faiz, Multan, Pakistan

Phone
+92 300 1234567

Email
info@stationerystore.com

Business Hours
Monday - Friday: 9:00 AM - 6:00 PM
Saturday: 10:00 AM - 4:00 PM

Send Us a Message

Full Name

Email Address

Subject

Message

Send Message

Stationery Store Home Products About **Contact** Admin Cart 0

Monday - Friday: 9:00 AM - 6:00 PM
Saturday: 10:00 AM - 4:00 PM

Message

Send Message

Stationery Store
Providing quality stationery since 2025

Quick Links
Home
Products
About
Contact

Contact Info
Air University, Chak-S Faiz, Multan, Pakistan
+92 300 1234567
info@stationerystore.com

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11. CRUD Operations Description

Operation	Explanation
Create	Add new stationery products
Read	Display all products
Update	Modify product details
Delete	Remove product

12. Testing & Results

12.1 Test Cases

Test Case	Expected Result	Status
Add Product	Product added	Passed
Update Product	Product updated	Passed
Delete Product	Product removed	Passed
Data Persistence	Data saved	Passed

13. Security & Data Validation

- Input validation to avoid empty fields
- Confirmation before delete action
- Controlled access to admin functions

14. Limitations

- No backend database
- No authentication system
- Local Storage dependency

15. Future Enhancements

- Backend integration (Node.js / PHP)
- Online payment gateway
- User accounts
- Search & filter functionality

16. Conclusion

The Stationery E-Commerce Website successfully demonstrates the practical implementation of ICT concepts and CRUD operations using front-end technologies. The project is suitable for academic and small business use

17. References

- W3Schools
- MDN Web Docs
- JavaScript Documentation
- Course Notes